

GetPawsy — Self-Healing Intelligence Layer (SHIL)

v1

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Architectural rule (permanent): every critical subsystem registers a probe → diagnose → recover → validate → learn contract. SHIL is the single brain that runs that contract on a 5-minute heartbeat. It does not replace Guardian, Commander, ACOS, OIE, ODE or Production Validation — it consumes them and acts on top.

Components now protected (16)

Pinterest Creative Factory, PCIE2 Publish Queue, missing pin images, Pinterest OAuth, Pinterest Cron Worker, Premium Engine pause flag, Checkout funnel collapse, Stripe session expiry burst, Analytics ingestion freshness, Frontend error rate, Pinterest pipeline failures, Cron freshness, Media asset 404, Cinematic worker heartbeat, Render worker heartbeat, Guardian status.

New tables (6)

shil_subsystems, shil_signatures, shil_playbooks, shil_incidents, shil_recoveries, shil_metrics_daily. All admin-only SELECT via has_role + service_role full access.

New edge functions (3)

self-healing-orchestrator (cron */5), self-healing-recoverer (allow-list playbook executor), self-healing-validator (re-runs probe and stamps validation_passed). All deployed.

Seeded playbooks (6)

restart_pinterest_cron_worker, replenish_creative_factory, requeue_pcie2_missing_images, refresh_pinterest_oauth, unpause_premium_engine_if_safe (preconditioned), escalate_only.

First live run (validation)

16 subsystems probed. 13 green, 1 yellow (pinterest_cron_worker idle 50 min), 2 red (pcie2_publish_queue with 120 stuck-ready pins; frontend_error_rate 72 errors/hour). 3 incidents opened. Recoveries dispatched: restart_pinterest_cron_worker → success (1.3s); requeue_pcie2_missing_images → in progress; frontend_error_rate → escalate_only.

Safety

SHIL never publishes, never charges, never alters security. Publishing/Stripe/security paths are escalate_only until promoted by recurring success.

Operator experience

/admin/self-healing — live subsystem grid, incidents feed, recovery history, learned signatures. Operators only notified for critical or unsafe-to-recover failures via guardian_notification_queue.

Success metrics tracked

MTTD, MTTR, auto-recovery rate, recurring-incident rate, false-positive rate, availability %. Rolled up into shil_metrics_daily.

Next iterations

Promote safe playbooks from escalate-only as confidence grows; per-subsystem upstream timestamps for accurate MTTD; nightly metrics rollup; Slack/email notification channel.